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Building a Basic Sentiment Analyzer with Python and TextBlob

ARTI 402: Programming for AI

Technical Report

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1. Introduction

In 2007, the iPhone was recognized by Time Magazine as the best invention of the year. This was rather expected, given that phones prior to the iPhone lacked seamless touchscreens and operating systems. Ever since, Apple has continually released new designs and models, most of the time building on and enhancing current models. In this way, Apple has managed to garner a vast and loyal band of customers, building amongst them anticipation for the next upcoming release every year.

2. Problem Statement

Recent releases of iPhone models have started to get repetitive, and this has not been dismissed by the countless critical product reviews of customers. It is important for Apple to make note of this, considering the company's reputation for revolutionizing modern technology. The objective of this analysis is to gauge the general sentiment regarding the release of the iPhone XR in India through Amazon reviews. The reason for selecting the iPhone XR was for its affordability compared to other iPhones at the time. The 'R' is believed to stand for 'reduced', meaning a more basic version of the iPhone X. As for the choice to base this analysis on the sales in India, it is interesting to note that the iOS market share is not very popular in South Asia. This intersection makes for a unique case study where analyzing the sales of a cheaper version of the iPhone in a place where the iPhone is not very popular will prove to be useful for Apple's sales performance in the region of South Asia as a whole.

3. AI Solution

The method used will be a machine-learning based approach to the sentiment analysis case study as opposed to the rule-based approach. We utulized TextBlob, a Python library built on NLTK, focused on performing natural language processing (NLP) tasks to solve problems regarding textual data. The TextBlob library uses the Naive Bayes algorithm for its classification

and extraction of sentiments. Along with that, we made use of the Pandas and Matplotlib libraries to properly manipulate and visualize the analyses of the data.

4. Case Study

As mentioned in the problem statement, the main focus of this study is the iPhone XR. We retrieved the dataset from Kaggle which was compiled by the Crawl Feeds web scraping service. Listed below are the column details of the original dataset.

Table 1: Dataset Column Metadata

Column name	Column description	Column data type
product	The product being reviewed	String
helpful_count	The number of people who found the review helpful	Integer
total_comments	The total number of comments on the review	Integer
url	The URL of the review post	String
review_country	The country from which the review was posted	String
reviewed_at	The date and time of the review submission	DateTime
review_text	The text of the review	String
review_rating	The rating given to the product by the reviewer	Integer
product_company	The company that manufactured the product	String
profile_name	The name of the reviewer	String
review_title	The title of the review	String

4.1 Data Description

The dataset consisted of over 5000 reviews on the iPhone XR in India. The next task prompted us to import the dataset into a Pandas DataFrame. We filtered the data to ensure the reviews were covering one product. We then used the standard `len()` method in addition to the `min()`, `max()`, and `mean()` methods to make sense of the dataset at hand. Our objective was to obtain the following information:

- Product name
- Number of rows
- Number of columns
- Length of the shortest review
- Length of the longest review
- Average length of the review

```
import pandas as pd
import matplotlib.pyplot as plt

product_names = df['product'].unique()
row_count = df.shape[0]
col_count = df.shape[1]
shortest_review = df['review_text'].str.len().min()
longest_review = df['review_text'].str.len().max()
average_review = df['review_text'].str.len().mean()

print("product name: " + product_names[0])
print("number of reviews: " + str(row_count))
print("number of columns: " + str(col_count))
print("shortest review length: " + str(shortest_review))
print("longest review length: " + str(longest_review))
print("average review length: " + str(average_review))
```

Figure 1: Dataset Description Implementation

With the implementation shown above, the description of our dataset came out as follows:

```
product name: Apple iPhone XR (64GB) - Black
number of reviews: 5010
number of columns: 12
shortest review length: 1.0
longest review length: 1001.0
average review length: 65.6145396444977
```

Figure 2: dataset description

4.2 Review Sentiments

After obtaining useful information regurding the dataset's metadata, the next step was to write and apply a function called `sentimentAnalyzer(text)` to the text column of our DataFrame. This creates a new column in the DataFrame called Sentiment, which includes the sentiment for each review. The coded implemntation of this can be found in Figure 3.

```
from textblob import TextBlob
from textblob.en import polarity


def sentimentAnalyzer(text):
    blob = TextBlob(text)
    if blob.sentiment.polarity > 0.2:
        return 'Positive'
    elif 0.2 >= blob.sentiment.polarity >= -0.2:
        return 'Neutral'
    else:
        return 'Negative'

df['review_text'] = df['review_text'].astype(str)
df['sentiment'] = df['review_text'].apply(sentimentAnalyzer)
```

Figure 3: Sentiment Analyzer Implementation

The code implementation in Figure 3 generated a table with an extra column that specifies the sentiment of each review based on its `review_text` column. There are issues with this regarding the true sentiment of the review due to field inconsistensies of reviews, however that will be covered later in this report (See section 4.4).

11 rows 1000 rows × 3 columns

	review_rating	review_text	sentiment
0	3.0 out of 5 stars	NOTE:	Neutral
1	1.0 out of 5 stars	Very bad experience with this iPhone xr phone....	Neutral
2	5.0 out of 5 stars	Amazing phone with amazing camera coming from ...	Positive
3	1.0 out of 5 stars	So I got the iPhone XR just today. The product...	Neutral
4	5.0 out of 5 stars	I've been an android user all my life until I ...	Positive
...	...	...	...
995	5.0 out of 5 stars	Best user experience till date.	Positive
996	5.0 out of 5 stars	Received genuine product with one year free sc...	Positive
997	5.0 out of 5 stars	it’s been a week since i got this and it’s bee...	Positive
998	5.0 out of 5 stars	I must say best iPhone device of apple. Colour...	Positive
999	5.0 out of 5 stars	Excellent product. If you are a fan of quality...	Positive

Figure 4: Sentiment Analyzer Column Results

4.3 Data Visualization

With the the newly generated sentiment table in Figure 4, the next step is the visualize this data into graphs and charts using Matlibplot. The first graph of interest was to see the relative count of the positive, neutral and negative reviews on the iPhone XR. This can be depicted through a simple bar graph as follows.

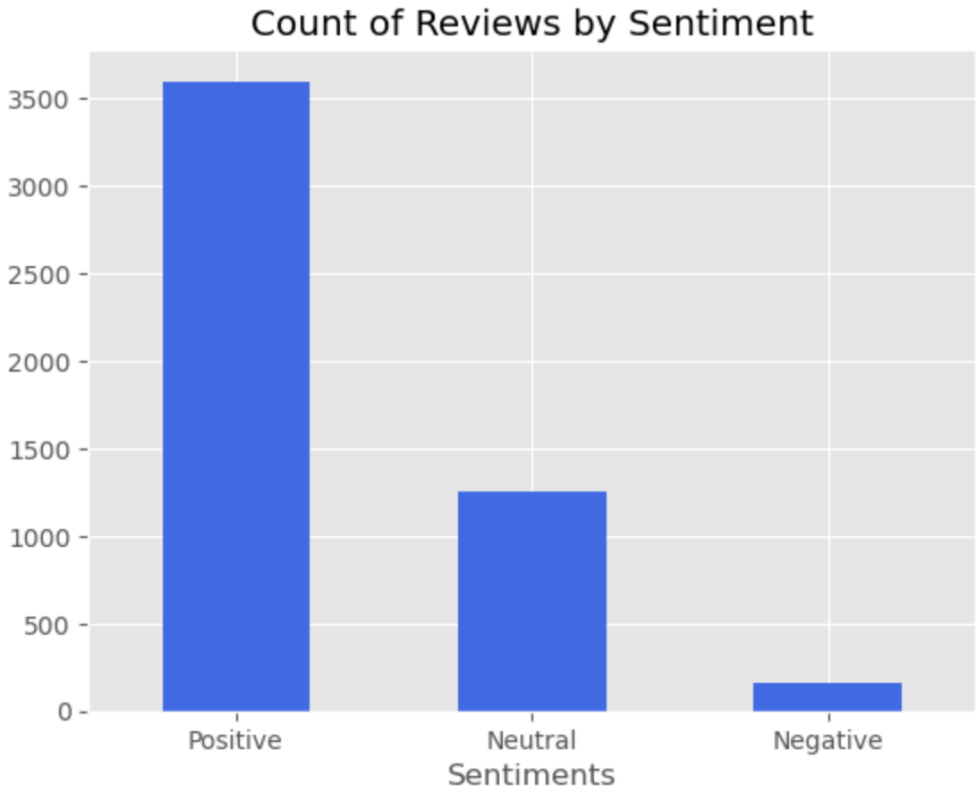


Figure 5: Count of Reviews by Sentiment

As we can see in Figure 5, the number of positive reviews is overwhelmingly high when compared to the neutral and negative reviws. The positive reviews go above 3500 in count, whereas the neutral reviews are between 1000 to 1500. The negative reviews are well under 500 in count, making them account for less than ten percent of the whole dataset. This shows a general posisitve sentiment towards the product.

The next step was to visualize the relative count of the reviews based on their star rating. This can give us an idea on how consistent the dataset is based on the count of posisitve sentiment reviews and those that are given a high star rating of four to five stars.

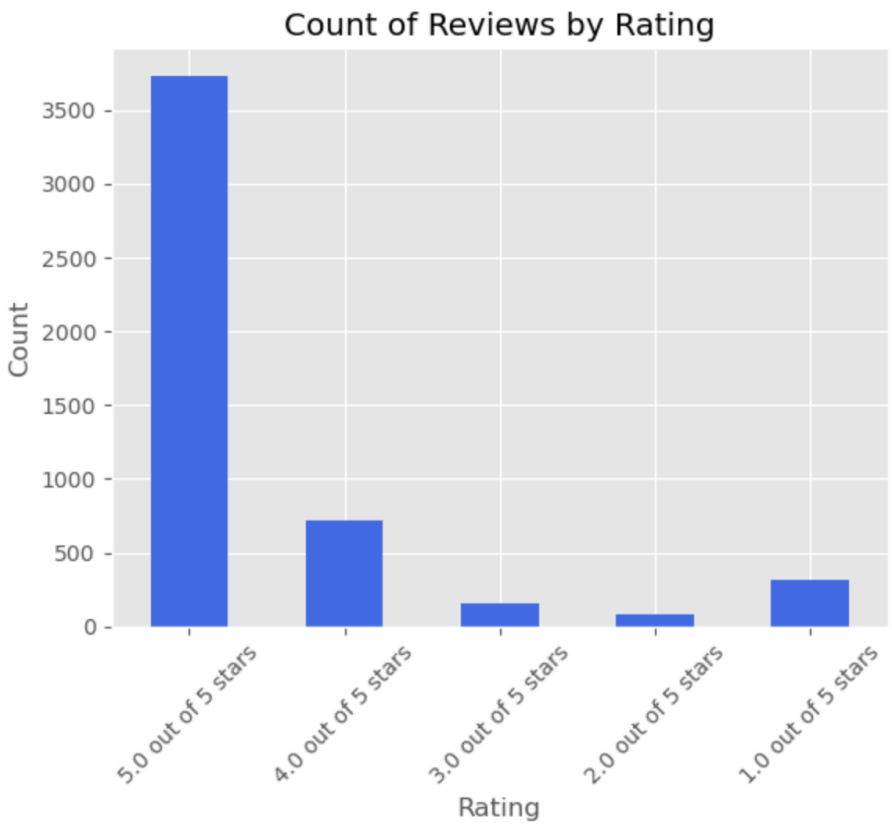


Figure 6: Count of Reviews by Rating

Figure 6 tells us that the dataset is fairly consistent, because there are over 3500 reviews with a five-star rating, and likewise there are over 3500 reviews that have a positive sentiment on the review-text alone. The two generated figures are useful in understanding the general picture. However, this leaves the scrutiny of information untouched, as it is also vital to understand where the outliers may exist.

This requires us to stretch the data range, and work on the **polarity** level. Polarity groups texts into levels of emotion that is being expressed. It is expressed as a numerical ration on the range [-1, 1]. Negative one being on the extreme negative end, and positive one being on the extreme positive end. Plotting this on a bar graph with our given dataset, we can see which polarity score exceeds its neighboring polarity increments in Figure 7.

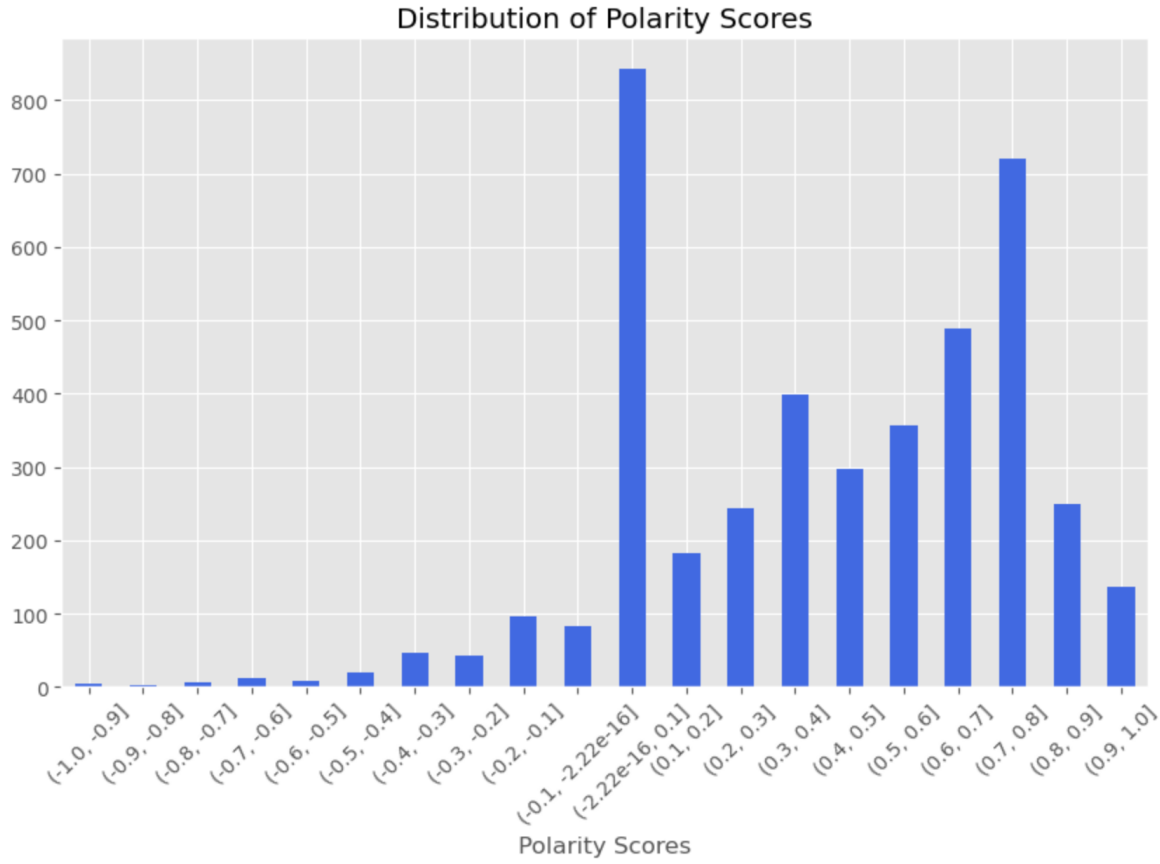


Figure 7: Distribution of Polarity Scores

The polarity range with the highest count is (-0.1, 0.0]. This accounts for the majority of the netural reviews, since those that fall under that sentiment must be in the range [-0.2, 0.2]. The second greatest polarity score is in the range (0.6, 0.7]. Increments before this gradually increase up to it, accounting for the overwhelming proportion of positive reviews to negative reviews. This can also be used as evidence to support the consistency of the dataset.

Another useful visualization is a bar graph of the most common words used in the reviews. While this may yield redundant information, it may be helpful in understanding the shared use of words among customers, in turn understanding the customers' priorities when reviewing the item. This can be seein in Figure 8.

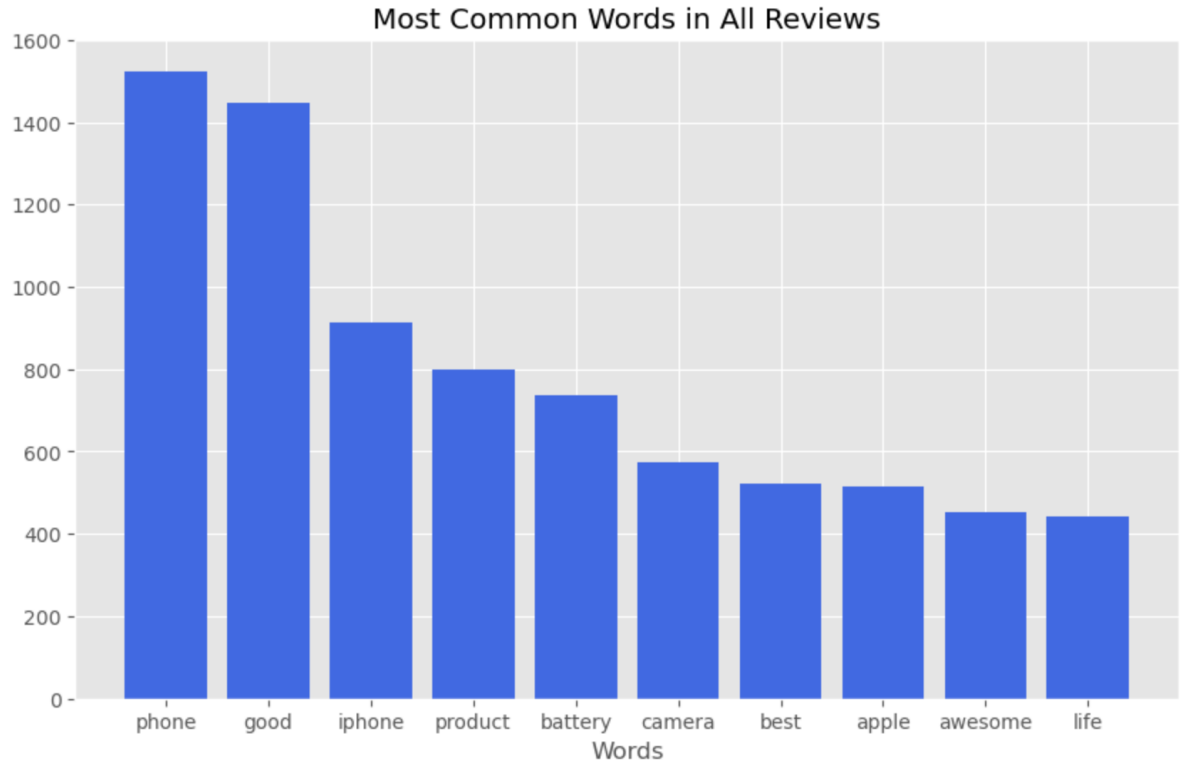


Figure 8: Most Common Words in All Reviews

Judging from this graph, we can see that the 'battery' and the 'camera' of the iPhone are shared topics of discussion on which customers regularly place an emphasis. This is important for Apple to make note of for their current marketing strategies and future products. Other parts of this graph prove to be redundant because of the nature of the dataset reviews.

4.4 Data Consistency

While the visualization section of this report prove there to be many useful depictions of the dataset, it is also likely that there are inconsistencies within the dataset. Glossing over the dataset, there does not seem to be anything out of place. However, when we delve deeper into the difference between review title and review text, or review text and star rating, we will notice that there exist gaps between these fields within individual reviews.

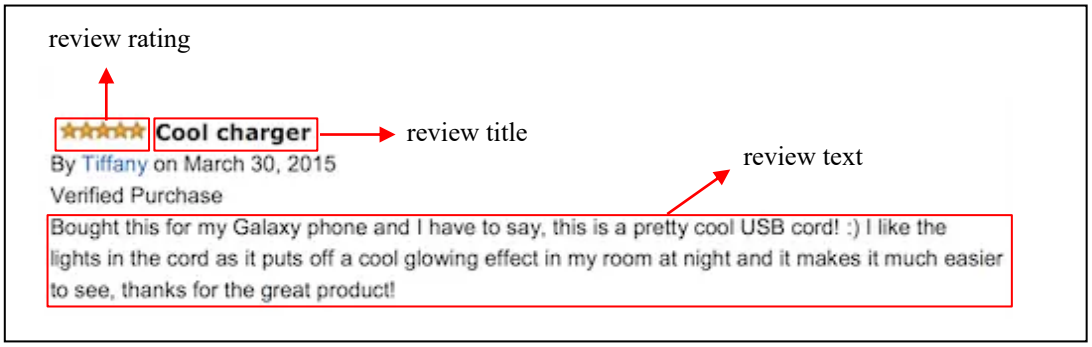


Figure 9: Amazon Review Fields Example

Above is an example of an Amazon review with the review rating, review title, and review text. Each of these have a sentiment of their own. In this section, we will investigate the difference between these fields, and see whether reviews are consistent for the most part. In Figure 10, the graph counts the number of consistent reviews based on their `review_rating` and `review_text`. We discarded the review title within this example for the sake of simplicity and accuracy.

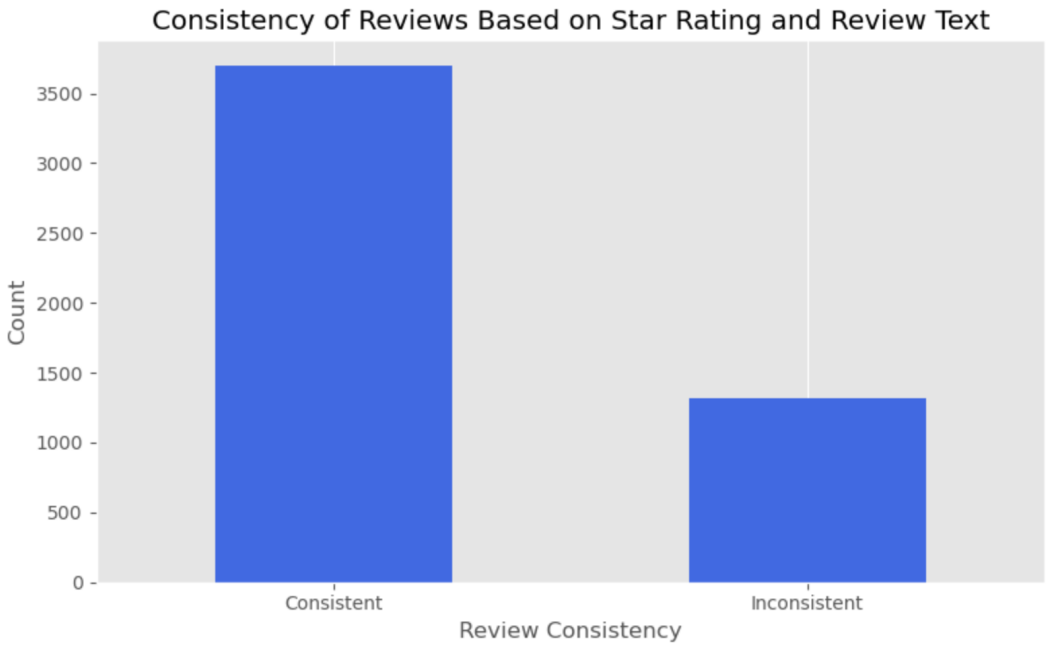


Figure 10: Consistency of Reviews Based on Star Rating and Review Text

In order to better understand how we obtained this statistic, we must reference the implementation and the numerical ranges used. For a review to count as consistent, the `review_rating` variable must align with the the `review_sentiment`. If the review star rating is four or higher, the polarity of the sentiment must accordingly be greater than 0.2, marking it as positive. If the review has a star rating of three, the review's sentiment polarity must fall between the range (-0.2, 0.2). Finally, if the review has two stars or less, the sentiment polarity of the review text must be -0.2 or lower.

```
def review_consistency(review_text, review_rating):

    review_rating = int(review_rating[0])
    review_sentiment = TextBlob(review_text).sentiment.polarity

    if review_rating == 5 or review_rating == 4:
        expected_sentiment = 0.2
    elif review_rating == 3:
        expected_sentiment = 0.0
    elif review_rating == 2 or review_rating == 1:
        expected_sentiment = -0.2
    else:
        return "Invalid star rating."

    if (review_sentiment >= expected_sentiment and review_rating >= 3) or
(review_sentiment < expected_sentiment and review_rating < 3):
        return "Consistent"
    else:
        return "Inconsistent"
```

Figure 11: Implementation to Check Dataset Consistency

The code above is a part of what yields the bar graph in Figure 10. The number of inconsistent reviews is much greater than previously expected. This is a very valuable statistic and can open doors for more investigation and insight as to why the reviews are inconsistent from the start. Some inconsistency stems from semantic errors in reviews. Spelling can account for a great deal of reviews that have an inconsistency between the star rating and the review text. This can also account for the review title. Figure 12 shows the difference in polarity between the `review_text` and `review_title` in each individual review.

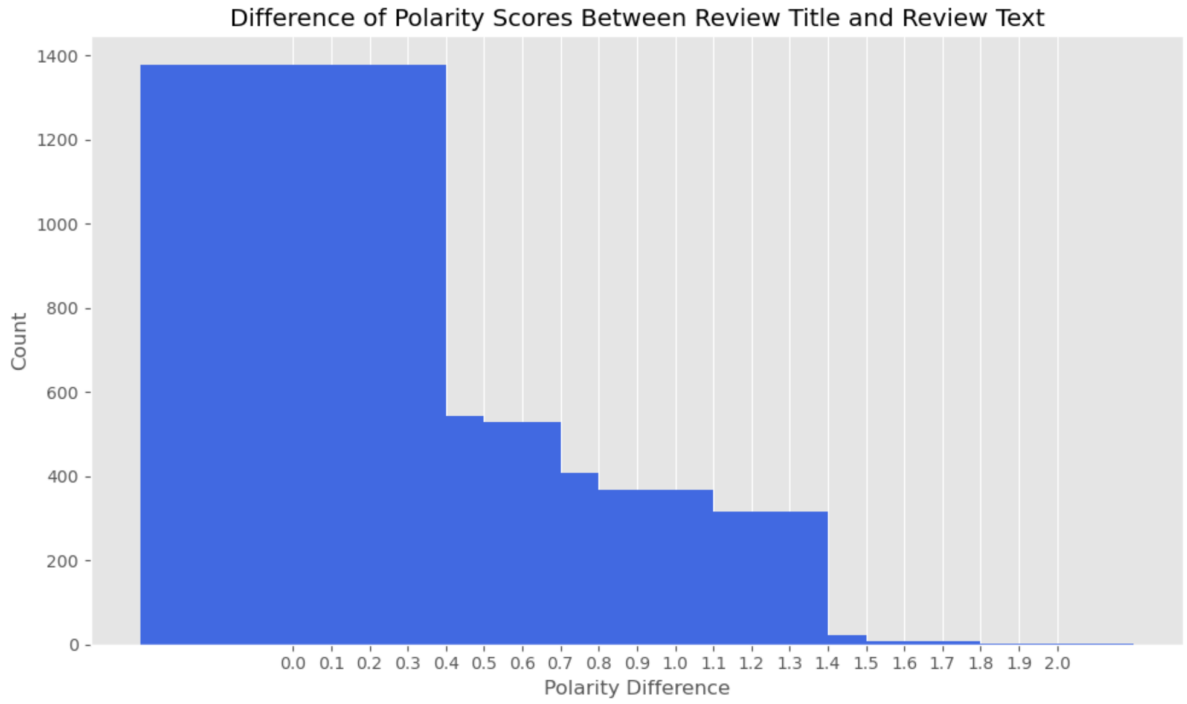


Figure 12: Difference of Polarity Scores Between Review Title and Review Text

Figure 12 is useful in allowing us to understand the extent of polarity score difference between the fields `review_text` and `review_title`. Over 1200 reviews have very little difference between these two fields. However, we must also make note of the accumalation of reviews that have a difference ranging between 0.4 to 1.4 in their polarity scores. There exist many reasons for this, and as was mentioned previously, this is due to the nature of the dataset.

There is nothing embedded in the Amazon review system to ensure the reviews that are sent in are syntactically and semantically apt. This accounts for many inconsistencies that data scientists will face when dealing with datasets like this one. That is why it is crucial to clean the dataset of spelling mistakes, punctuation mistakes, rhetorical nuances and other exceptions, allowing for the results of its analysis to be useful. Figure 13 shows one example of a spelling mistake which affected the polarity score difference drastically between the review text and review title, deeming it inconsistent.

61-70 1000 rows × 3 columns

	review_title	review_text	polarity_difference
60	I love it..!	Awesome IPhone.. I have upgraded from iPhone 6 and I can noti…	0.2
61	Xcellent phone	Got this phone for offer price. Everything is great from pack…	0.9
62	Best iPhone to buy at this price.	Bought this phone for my wife. She's absolutely happy with th…	0.2
63	Swap my iphone xr	I want to replace this iphone because since i got this it wor…	0.1
64	Thanks apple India. Very very thanks amazo…	Great phone for better photos and low light battery usage is …	0.2
65	Good phone but horribly heavy	So overall it's a pretty nice phone, I honestly don't find mu…	0.0
66	Very good phone very fast	Good phone	0.1
67	My First Apple iPhone	I've been using Android for close to a decade and now decided…	0.4
68	Nice Phone	Using this phone since 2 months. Here is my review. Hope you …	0.3
69	XR Deals seem very good	I was skeptical looking at some XR reviews and also purchasin…	0.7

Figure 13: Semantic Error in Review Example

5. AI Solution to Client's Needs

5.1 Sentiment Analysis with TextBlob

TextBlob is a powerful Python library designed for processing textual data, particularly useful in the field of natural language processing (NLP). With its built-in sentiment analysis feature, TextBlob computes polarity and subjectivity scores, allowing us to gauge the emotional tone of a piece of text. The polarity score ranges from -1 to 1, the negative end having a bad sentiment and the positive end having a positive sentiment, making it easy to classify texts based on their emotional content. TextBlob was an essential AI feature in this case study due to its ease of use and efficient applicability. Without this library, it would have been much more challenging to obtain the sentiment of a review, which proved to be fundamental to this analysis. This functionality is not only invaluable in assessing customer feedback, but further enhances our understanding as well as that of our clients'. Having this kind of data-driven and sentimental insight to their product, are clients are better equipped to make stratigic decisions moving forward.

5.2 Data Manipulation with Pandas and Numpy

Pandas and Numpy are essential libraries for data manipulation and numerical computations in Python, making them integral to sentiment analysis workflows and machine learning applications

in general. Pandas provides the use of DataFrames that allow for efficient organization and manipulation of large datasets. Numpy complements this by offering powerful mathematical functions and operations on arrays, enabling efficient computations on numerical data. Together, they facilitate the extraction and preparation of sentiment-related features, as was seen throughout this report. Using these two libraries, we were able to calculate the average sentiment scores across different categories and analyze trends within our dataset. This further supported our analytical investigation of the overal sentiment of our client's customers.

5.3 Visualization with Matplotlib

Matplotlib is a widely-used library for data visualization in Python, providing the tools necessary to create informative and easy-to-read plots. In the context of this report, visualizing sentiment trends, distributions, and comparisons yielded actionable insights. For example, by plotting the frequency of sentiment scores or the distribution of positive versus negative reviews, we were able to identify patterns and outliers in the data, and make note of this to our client. By integrating Matplotlib with libraries like Pandas and TextBlob, we were able to effectively communicate the findings of the sentiment analysis, making it easier for our client to understand and act on the insights derived from textual data.

6. Conclusion

In summary, the combination of TextBlob, Pandas, Numpy, and Matplotlib formed a robust toolkit for conducting the sentiment analysis seen in this report. TextBlob streamlined the extraction of customer sentiments from Amazon reviews, while Pandas and Numpy provided the necessary infrastructure to manipulate Amazon reviews into something statistically useful. Additionaly, Matplotlib was utilized to enhance the interpretability of our sentiment analysis through clear visualizations. Together, these libraries yielded valuable insights from what started as a csv file containing Amazon reviews. This report now enables data-driven decision-making and a deeper understanding of public sentiment for our client.